

BREAKING DATA EFFICIENCY DILEMMA: A FEDERATED AND AUGMENTED LEARNING FRAMEWORK FOR ALZHEIMER'S DISEASE DETECTION VIA SPEECH

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ABSTRACT

Early diagnosis of Alzheimer's Disease (AD) is crucial for delaying its progression. While AI-based speech detection is non-invasive and cost-effective, it faces a critical data efficiency dilemma due to medical data scarcity and privacy barriers. Therefore, we propose FAL-AD, a novel framework that synergistically integrates federated learning with data augmentation to systematically optimize data efficiency. Our approach delivers three key breakthroughs: First, absolute efficiency improvement through voice conversion-based augmentation, which generates diverse pathological speech samples via cross-category voice-content recombination. Second, collaborative efficiency breakthrough via an adaptive federated learning paradigm, maximizing cross-institutional benefits under privacy constraints. Finally, representational efficiency optimization by an attentive cross-modal fusion model, which achieves fine-grained word-level alignment and acoustic-textual interaction. Evaluated on ADReSSo, FAL-AD achieves a state-of-the-art multi-modal accuracy of 91.52%, outperforming all centralized baselines and demonstrating a practical solution to the data efficiency dilemma. Our source code is publicly available at <https://github.com/smileix/fal-ad>.

Index Terms— Alzheimer's Disease Detection, Federated Learning, Data Augmentation, Voice Conversion, Speech and Language Analysis

1. INTRODUCTION

Alzheimer's Disease (AD) is a prevalent neurodegenerative disease that severely impacts patients' memory, cognition and behavior, posing a substantial burden on global public health systems. Due to its irreversible nature and prolonged preclinical stage, early diagnosis is critical for delaying disease progression and improving patients' quality of life [1]. In recent years, owing to the non-invasiveness, low cost and scalability of spontaneous speech analysis, it has emerged as a promising tool for AD screening [2], where deep learning methods can capture subtle signs of cognitive decline, demonstrating significant potential for practical application.

High-performance deep learning diagnostic models relies heavily on large-scale, high-quality annotated speech data, however, the lack of such data is a fundamental bottleneck in current research. This bottleneck reflects a data efficiency dilemma in medical AI, the

inherent data scarcity and privacy-related barriers. This challenge manifests at two interrelated levels: first, the absolute lack of efficiency, e.g., data scarcity, collecting AD speech data requires rigorous clinical coordination, resulting in high costs and long cycles, which inherently limits the amount of available data. This seriously constrains the model's learning capacity, leading to overfitting and weak generalization [3]. Second, the relative efficiency loss, e.g., data silos, due to medical privacy regulations and institutional barriers, already scarce data are further fragmented across hospitals and research centers, preventing collaborative integration and scale benefits [4]. The two issues reinforce each other, creating a vicious cycle.

Existing studies have yet to systematically address this dilemma. Most mainstream efforts focus on designing more complex models (e.g., [5, 6]) to extract more discriminative features, or exploring multi-modal fusion to capture cross-modal interactions (e.g., [7, 8]), yet largely overlook the fundamental issue of data efficiency dilemma. Although recent work such as [9, 10, 11] has begun to explore federated learning (FL) to mitigate privacy barriers, these approaches only establish a basic collaborative framework without resolving the inherent problem of data scarcity. Moreover, most methods adopt simple late fusion strategies, failing to fully exploit the representational capacity of the data. Another study [12] also highlights that standard FL may under-perform in highly heterogeneous and scarce data environments. None of these methods offers a comprehensive solution to enhance data utilization efficacy.

To fundamentally address the data efficiency bottleneck, we propose the Federated and Augmented Learning framework for Alzheimer's Disease Detection (**FAL-AD**), whose core idea is to comprehensively optimize the data efficiency from generation to utilization. To our knowledge, this is the first work to systematically explore data efficiency. Our contributions are threefold:

- Absolute efficiency improvement: we construct a cross-category voice conversion-based data augmentation strategy to generate high-quality pathological samples, effectively expanding the data volume and diversity.
- Collaborative efficiency breakthrough: we design an adaptive federated learning paradigm that establishes a cross-silo collaboration mechanism, connecting isolated data silos into an efficient cooperative network.
- Representational efficiency optimization: we improve an attention-based cross-modal fusion model to achieve fine-grained alignment and deep interaction, ensuring maximal knowledge acquiring from limited data.

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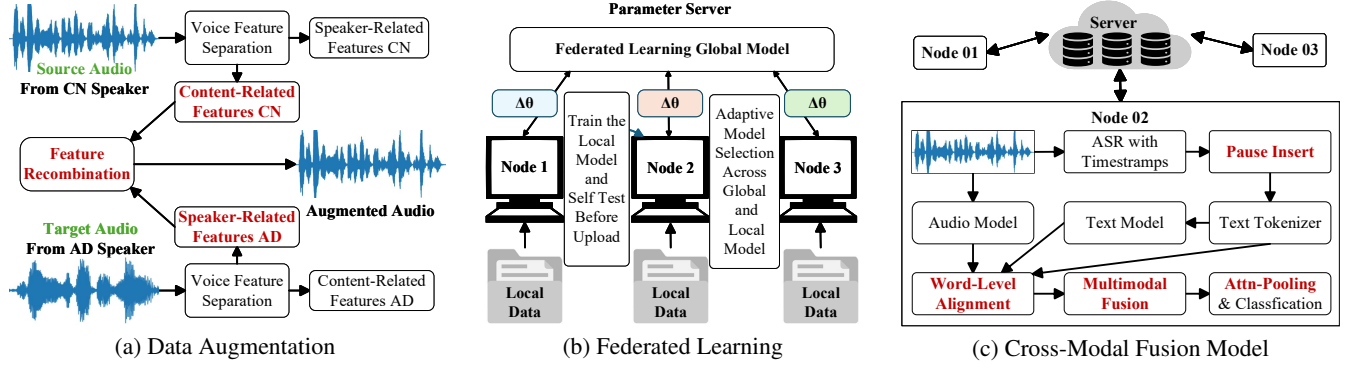


Fig. 1. Overview of the proposed FAL-AD framework. (a) Cross-category Voice Conversion-based Data Augmentation: generates pathological speech samples by recombining speaker features from one category and content features from another. (b) Adaptive Federated Learning: clients collaboratively train under privacy constraints and select their optimal model from the federation history. (c) Cross-Modal Fusion Model: depicts the word-level alignment and attentive fusion process of acoustic and textual features.

2. THE FEDERATED AND AUGMENTED FRAMEWORK

2.1. Overview

Fig.1 illustrates the overall framework of our proposed FAL-AD, which consists of three core modules: (1) a data augmentation module, responsible for expanding the training data; (2) a federated learning module, used for collaborative model training; and (3) a cross-modal fusion module, used for final classification decision. Each module is described in detail below.

2.2. Data Augmentation with Cross-category Voice Conversion

To address the scarcity of AD speech data, we employ a voice data augmentation strategy. While text-to-speech (TTS) synthesis is commonly used for augmentation, TTS systems are typically trained on healthy speech samples and offer limited controllability. Even after adjusting parameters such as speech rate and pause duration, the synthesized speech inevitably exhibits artificial characteristics and fails to capture the authentic pathological patterns of real AD patients. To more effectively generate disease-relevant vocal manifestations while minimizing interference from non-pathological factors (e.g., speaker identity), we introduce voice conversion (VC) technology, as shown in Fig.1(a). This method preserves the linguistic and pathological content of speech while transforming speaker-dependent characteristics, thereby enabling the model to focus on clinically relevant features.

Specifically, Our voice conversion is implemented using the CosyVoice 2.0B model [13]. Let $V_i = (S_i, C_i, Y_i)$ denote an original speech sample, where $Y_i \in \{AD, CN\}$ (CN stands for cognitively normal), S_i denotes the speaker characteristics (including timbre and identity-related features), and C_i represents the content aspects (including linguistic content, rhythm, pauses, and pathology-related acoustic features). During augmentation, for source sample V_i and target sample V_j with the opposite category ($Y_i \neq Y_j$), we employ voice conversion to recombine the speaker characteristics of the target audio S_j with the content features of the source audio C_i , thereby generating a new augmented sample:

$$\tilde{V}_{i,j} = \mathcal{VC}(V_i, V_j) = (S_j, C_i, Y_i), \quad (1)$$

where $\mathcal{VC}(\cdot)$ denotes the voice conversion function, $Y_i \neq Y_j$, and the label of the augmented sample satisfies $\tilde{Y}_{i,j} = Y_i$. Each sample in the dataset is alternately treated as the target sample, while a sample

from the opposite category is randomly selected as the source sample to perform the aforementioned voice conversion.

Through cross-category voice conversion, we generate both positive and negative class samples for each speaker, significantly enhancing the dataset's diversity. This strategy also resolves class imbalance issues while ensuring models no longer favor majority classes. Crucially, by maintaining identical speaker timbre across both sample types, the technique forces models to concentrate exclusively on learning pathology-related acoustic features, such as AD-specific prosodic abnormalities and pause patterns, and reduces the risk of model overfitting to irrelevant speaker-specific features. This innovative decoupling of timbre and pathological traits not only ensures precise control over diagnostic characteristics but also substantially improves the model's accuracy, robustness, and generalization capability to unseen samples.

2.3. Federated Learning with Adaptive Model Selection

We adopt Federated Learning (FL) as the foundational paradigm for collaborative training, as Fig.1(b) shows. In the standard Federated Averaging (FedAvg) procedure [14], the server executes the following steps at each communication round r :

1. **Server Distribution:** The central server distributes the current global model parameters ω^r to all participating clients.
2. **Local Client Update:** Each client i performs local training for E epochs using its local dataset $\mathcal{D}_i$, resulting in a set of local model updates $\Delta\omega_i^r$.
3. **Model Aggregation:** The server aggregates the received model updates from the clients:

$$\omega^{r+1} \leftarrow \omega^r + \sum_{i=1}^M \frac{n_i}{n} \Delta\omega_i^r, \quad (2)$$

where $n = \sum_{i=1}^M n_i$ denotes the total number of training samples across all clients, and n_i is the number of samples on client i .

However, the standard FL framework solely outputs the final aggregated global model ω^G , potentially overlooking superior intermediate models obtained during training. To mitigate this limitation, we introduce an **adaptive model selection strategy**, which operates as follows:

1. **Performance Tracking:** In each federated round r , after completing local training and before uploading the model to the server, each client i evaluates the performance of its local model ω_i^r on a local validation set and retains a copy of the model. Concurrently, after the server completes global aggregation and distributes the new global model ω^{r+1} , each client downloads this global model and evaluates the performance of the global model on the same local validation set, also saving the result.
2. **Optimal Model Selection:** This process repeats for a pre-defined number of communication rounds (e.g., $R = 30$ rounds). Upon conclusion of the federated training, each client i selects the model with the best performance on its local validation set from all saved model snapshots as its ultimate deployment model, denoted ω_i^* .

Our adaptive strategy maximizes performance and adaptability by dynamically selecting the optimal model ω_i^* for each client, whether a personalized local update or a generalized global model. The approach remains simple and efficient, introducing no extra hyper-parameters and adding only minimal evaluation and storage overhead. Each client thus deploys a tailored model ω_i^* that ensures high performance and local relevance.

2.4. Attentive Cross-Modal Fusion Model

To effectively integrate information from speech and text modalities, we build upon the CogniAlign [15] framework, as shown in Fig.1(c), where we conduct targeted improvements to adapt it for the federated learning environment. The final fusion model achieves fine-grained multimodal interaction through the following mechanisms:

First, as for feature extraction and alignment, we begin with prosodic information modeling, where the Whisper [16] automatic speech recognition (ASR) model generates transcriptions with word-level timestamps. According to the timestamps, three types of pause markers (comma, period, and ellipsis) are inserted into the text sequence to capture pause and disfluency patterns [17]. Subsequently, we use pre-trained models to extract token-level textual features and frame-level acoustic features, and then according to the timestamps, the latter is mean-pooled into word level to complete alignment.

Second, as for the feature fusion, we employ a gated cross-attention mechanism, implemented through a single-layer Transformer encoder to enable deep interaction between modalities [18]. Specifically, the audio embeddings serve as the Query while the text embeddings provide the Key and Value. The gated cross-modal attention is computed as:

$$\mathbf{H}_{\text{att}} = \text{Attention}(\mathbf{A}, \mathbf{T}, \mathbf{T}), \quad \mathbf{G} = \sigma(\mathbf{W}_g \mathbf{H}_{\text{att}} + \mathbf{b}_g), \quad (3)$$

$$\mathbf{H} = \mathbf{G} \odot \mathbf{H}_{\text{att}} + (1 - \mathbf{G}) \odot \mathbf{A}, \quad (4)$$

where $\mathbf{A}$ and $\mathbf{T}$ respectively represent the word-level embedding sequences of audio and text, σ denotes the sigmoid function, $\mathbf{W}_g$ and $\mathbf{b}_g$ are trainable parameters, and $\odot$ represents element-wise multiplication. And then an attention pooling layer dynamically learns the importance of each token through the learned weights:

$$\alpha_i = \mathbf{W}_a \mathbf{H}_i + \mathbf{b}_a, \quad w_i = \text{softmax}(\alpha)_i, \quad (5)$$

$$\mathbf{h} = \sum_{i=1}^T w_i \mathbf{H}_i, \quad (6)$$

where $\mathbf{W}_a$ and $\mathbf{b}_a$ are trainable parameters. The final classification is performed by a multilayer perceptron (MLP) on the aggregated representation $\mathbf{h}$.

3. EXPERIMENTS

3.1. Dataset and Preprocessing

We experiment with the ADReSSo Challenge dataset [19], which contains spontaneous speech recordings from 237 subjects (118 AD and 119 CN), describing the "Cookie Theft" picture. To ensure a fair comparison with existing works [15], we adopt a five-fold cross-validation strategy. The dataset is randomly partitioned into five mutually exclusive folds, iteratively using one fold as the test set and the remaining four as the training set. The final reported performance metrics are the average results across all five folds. Note that only the training set is augmented. For both local and federated paradigms, the training set is further randomly divided into three parts to serve as the local data for three clients.

3.2. Experimental Setup

To comprehensively evaluate the efficacy of the FAL-AD framework, we designed multiple comparative experiments from the perspective of learning paradigms, systematically measuring the trade-off between data collaboration and privacy preservation. Under each learning paradigm, model variants with three input modalities, i.e., audio, text, and multi-modal, so as to thoroughly validate the performance of different modalities under various paradigms.

In machine learning, Centralized Learning (CL) represents the ideal scenario of data sharing, and serves as the reference for the performance upper bound. Local Learning (LL) simulates the complete data silo scenario, where each client trains independently using only its local data without any collaboration, representing the performance lower bound. Federated Learning (FL) enables collaboration among clients while keeping data localized. This constitutes a rigorous comparative system. Comparing CL with LL quantifies the performance degradation caused by data silos; comparing LL with FL validates the fundamental benefits of federated collaboration; comparing standard FL with our proposed FAL-AD verifies the added value of our data augmentation and adaptive federated strategies.

3.3. Implementation Details

All audios are transcribed using the Whisper-large-v3 model [16] to obtain word-level timestamps. The audio is down-sampled to 16kHz. Audio and text features are respectively extracted using the frozen pre-trained models facebook/wav2vec2-base-960h [20] and distilbert-base-uncased [21], with the maximum sequence length set to 200. The multimodal fusion module employs with a hidden dimension of 768 and 12 attention heads. The classifier is a two-layer MLP. The hyper-parameter settings for CL and LL baselines are consistent with CogniAlign. Our FL methods are trained using the AdamW optimizer with a learning rate of $5e-5$, weight decay of 0.01, and a fixed batch size of 64, which simulates 3 clients for 30 communication rounds. All hyper-parameters are selected through grid search to ensure optimal model performance.

3.4. Results and Analysis

Overall, our proposed FAL-AD framework achieves state-of-the-art performance on the ADReSSo dataset, as shown in Table 1, surpassing all existing centralized benchmarks. The experimental results systematically validate its effectiveness across three key aspects: first, the comparison between CL and LL quantifies the severe performance degradation caused by data isolation, highlighting the necessity of collaborative paradigms. Second, FL effec-

Table 1. Performance comparison (Accuracy and F1-Score in %) across different modalities and learning paradigms on the ADReSSo dataset. Results from previous centralized methods are compared against our implementations under Centralized Learning (CL), Local Learning (LL), and Federated Learning (FL) paradigms, both with and without data augmentation (Aug). The best value in each group is marked with an underline, and the global best value across all groups is marked with bold. Note that CL here is the strict reproduction version of CogniAlign using its source code.

Methods → Modals ↓		Previous Work (Centralized Paradigm)				Our Work					
		C-Attn [22]	Ying [23]	Bang [24]	CogniAlign [15]	CL	CL+Aug	LL	LL+Aug	FL	FL+Aug
Audio	Acc	75.30	71.20	69.01	<u>80.12</u>	74.55	<u>79.39</u>	<u>68.69</u>	68.08	83.84	85.05
	F1	76.00	73.10	70.39	<u>79.46</u>	73.39	<u>79.14</u>	65.68	<u>67.10</u>	83.67	84.64
Text	Acc	73.50	78.90	83.10	<u>86.77</u>	84.85	<u>86.67</u>	78.39	<u>79.80</u>	87.68	90.30
	F1	73.50	79.00	83.10	<u>86.59</u>	84.69	<u>86.63</u>	77.48	<u>79.55</u>	87.64	90.28
Both	Acc	77.20	83.70	87.32	<u>90.36</u>	86.06	<u>86.67</u>	78.59	<u>80.61</u>	89.70	91.52
	F1	77.60	83.30	87.25	<u>90.11</u>	85.89	<u>86.64</u>	77.16	<u>80.35</u>	89.65	91.45

Table 2. Performance comparison (Accuracy and F1-Score in %) of federated learning algorithms with different model selection strategy: standard (sFL), personalized (pFL), and adaptive (aFL) federated learning. The best performance across section strategies is bold, while the best performance across FL algorithms is underlined.

Algorithms	sFL		pFL		aFL	
	Acc	F1	Acc	F1	Acc	F1
FedAdam	<u>90.91</u>	<u>90.88</u>	90.00	89.97	90.91	90.88
FedAdagrad	89.70	89.58	88.79	88.74	90.00	89.95
FedYogi	87.27	87.26	88.48	88.40	88.79	88.71
FedProx	89.70	89.63	90.60	90.52	90.91	90.84
FedAvg	<u>90.91</u>	90.85	<u>90.81</u>	<u>90.75</u>	91.52	91.45

tively mitigates this issue, achieving a performance gain of over 10 percentage points compared to isolated training (LL), demonstrating its capability to break down data silos while preserving privacy. Finally, and most significantly, our FL+Aug approach not only matches but exceeds the performance of its centralized counterpart (CL+Aug), achieving a new multi-modal accuracy benchmark of 91.52% and outperforming previous state-of-the-art methods including CogniAlign (90.36%). This paradigm breakthrough reveals a synergistic effect: federated aggregation acts as a regularizer to reduce overfitting, while data augmentation enhances generalization through expanded diversity.

Analysis on Data Augmentation: We further analyze the utility of the proposed voice conversion strategy under different learning paradigms. This strategy brings consistent performance improvements in almost all settings. The benefit is most prominent under the Federated Learning (FL) paradigm, boosting multimodal accuracy from 89.70% to 91.52%. This indicates that data augmentation provides richer and more diverse local data ‘fuel’ for federated learning, synergizing with the regularization effect of federated aggregation to push model performance to new heights. The gains under the Local Learning (LL) paradigm are also significant (multimodal accuracy increases from 78.59% to 80.61%), confirming its value as an effective regularization in extremely data-scarce environments. Notably, the improvement in CL paradigm is relatively limited (+0.61%), consistent with our previous finding that models trained on sufficient data are more susceptible to overfitting to the subtle biases introduced by augmented data. This contrast, in turn, demonstrates that our federated learning framework is a superior environment for de-

ploying such data augmentation strategies.

Analysis on Federated Learning: The performance of our FAL-AD surpasses the strong centralized baselines. This may be because that, while centralized training has access to all data, it can easily lead the model to overfit to specific patterns in the training set. The inherent ‘local update & global average’ iterative mechanism in the FedAvg algorithm effectively constrains the model optimization path, forcing it to converge to a flatter and more robust optimum for all data distributions, thereby achieving superior generalization performance. In addition, the adaptive model selection strategy ensures each client finally deploys its historical best model, maximizing the potential of models from different communication rounds. As shown in Table 2, while different aggregators exhibit varying affinities for not identically and independently distributed (non-IID) data, our proposed adaptive Federated strategy (aFL) demonstrates significant and consistent advantages. It achieves optimal performance under most configurations, validating its effectiveness as a lightweight, hyperparameter-free personalization solution. Notably, traditional local fine-tuning strategies (pFL) did not bring stable gains, suggesting that mandatory local fine-tuning in heterogeneous data environments may lead the model to deviate from the optimal solution. This proves that aFL possesses strong fault tolerance and stability. The dynamic selection mechanism retains the deployment simplicity of standard FL (sFL) while achieving personalization benefits close to pFL.

4. CONCLUSION

This paper presents FAL-AD, a novel framework that tackles the data efficiency dilemma in Alzheimer’s disease detection through federated learning and voice conversion-based augmentation. Our approach systematically addresses data scarcity and privacy constraints by enhancing absolute data efficiency through cross-category voice recombination, improving collaborative efficiency via adaptive federated learning with personalized model selection, and optimizing representational efficiency using attentive cross-modal fusion. Experimental results demonstrate that FAL-AD not only overcomes the limitations of data silos but achieves state-of-the-art performance with 91.52% accuracy, surpassing existing centralized baselines while maintaining privacy preservation. In the future, we plan to explore more advanced generative augmentation techniques and sophisticated personalization methods for non-IID data scenarios, to validate the framework’s generalization across multi-institutional datasets.

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